

LAPORAN PRAKTIKUM

Mata Kuliah Basis Data



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MODUL 13

JURNAL

Soal :

OUTPUT: SCREENSHOOT LAYAR HASIL RUNNING QUERY MONGOSHELL

Disediakan dataset berikut:

```
db.toko.insertMany([
{
"name" : "Toko Berkah",
"category" : "Elektronik",
"city" : "Jakarta",
"address" : { "street" : "Jl. Sudirman", "zipcode" : "10220" },
"products" : [
{ "item" : "TV", "price" : 3500000, "stock" : 10 },
{ "item" : "Kulkas", "price" : 2800000, "stock" : 5 }
]
},
{
"name" : "Toko Maju",
"category" : "Elektronik",
"city" : "Bandung",
"address" : { "street" : "Jl. Asia Afrika", "zipcode" : "40111" },
"products" : [
{ "item" : "HP", "price" : 1500000, "stock" : 20 },
{ "item" : "Laptop", "price" : 8000000, "stock" : 3 }
]
},
{
"name" : "Toko Salim Bakery",
"category" : "Elektronik",
"city" : "Jakarta",
"address" : { "street" : "Jl. Sudirman", "zipcode" : "10220" },
"products" : [
{ "item" : "Monitor LED", "price" : 7700000, "stock" : 11 },
{ "item" : "Airfryer Panasonic", "price" : 4850000, "stock" : 7 }
]
},
{
"name" : "Toko Sudinotech",
"category" : "Elektronik",
"city" : "Surabaya",
"address" : { "street" : "Jl. Sudirman", "zipcode" : "10220" },
"products" : [
```

```
{ "item" : "RTX 55000", "price" : 8500000, "stock" : 20 },
{ "item" : "Set PC Gaming Radeon", "price" : 12000000, "stock" : 9 }
],
},
])
```

1.) Tambahkan 1 toko baru bebas ke koleksi toko, dengan ketentuan:

- Minimal memiliki 2 produk di dalam array products
- Isi semua field: name, category, city, address, products

Hint:

```
db.toko.insertOne({
  "name": "...",
  "category": "...",
  "city": "...",
  "address": { "street": "...", "zipcode": "..." },
  "products": [
    { "item": "...", "price": ..., "stock": ... },
    { "item": "...", "price": ..., "stock": ... }
  ]
})
```

2.) Tampilkan semua toko yang berada di kota Jakarta.

3.) Tampilkan toko yang memiliki produk dengan price lebih dari 3.000.000.

Hint:

- Gunakan dot notation: products.price
- Gunakan operator: \$gt

4.) Ubah category Toko Berkah dari Fashion menjadi Aksesoris.

Tambahkan 1 produk baru ke dalam array products milik Toko Maju menggunakan \$push.

Hint:

- Filter berdasarkan name: Toko Maju
- Gunakan operator \$push untuk menambah elemen ke array
- Hitung jumlah toko per city, urutkan dari kota yang paling banyak tokonya.

Gunakan:

- \$group dengan \$sum: 1
- \$sort descending (-1)

6.) Hitung rata-rata harga produk per category.

Tahapan pipeline:

- \$unwind: '\$products' (bongkar array products dulu)
- \$group: kelompokkan per category, hitung \$avg dari products.price
- \$sort: urutkan sesuai kebutuhan

7.) Tambahkan 1 produk baru ke dalam array products milik Toko Maju menggunakan \$push.

Hint:

- Filter berdasarkan name: Toko Maju
- Gunakan operator \$push untuk menambah elemen ke array

8.) Tampilkan harga produk tertinggi (\$max) per city.

Hint:

- Unwind products terlebih dahulu
- Gunakan \$max pada products.price

9.) Buat pipeline lengkap dengan 4 tahap:

- \$match — filter hanya toko category Elektronik
- \$unwind — bongkar array products
- \$group — hitung total stock per city
- \$sort — urutkan dari total stock terbesar

10.) Hapus toko yang memenuhi dua kondisi sekaligus:

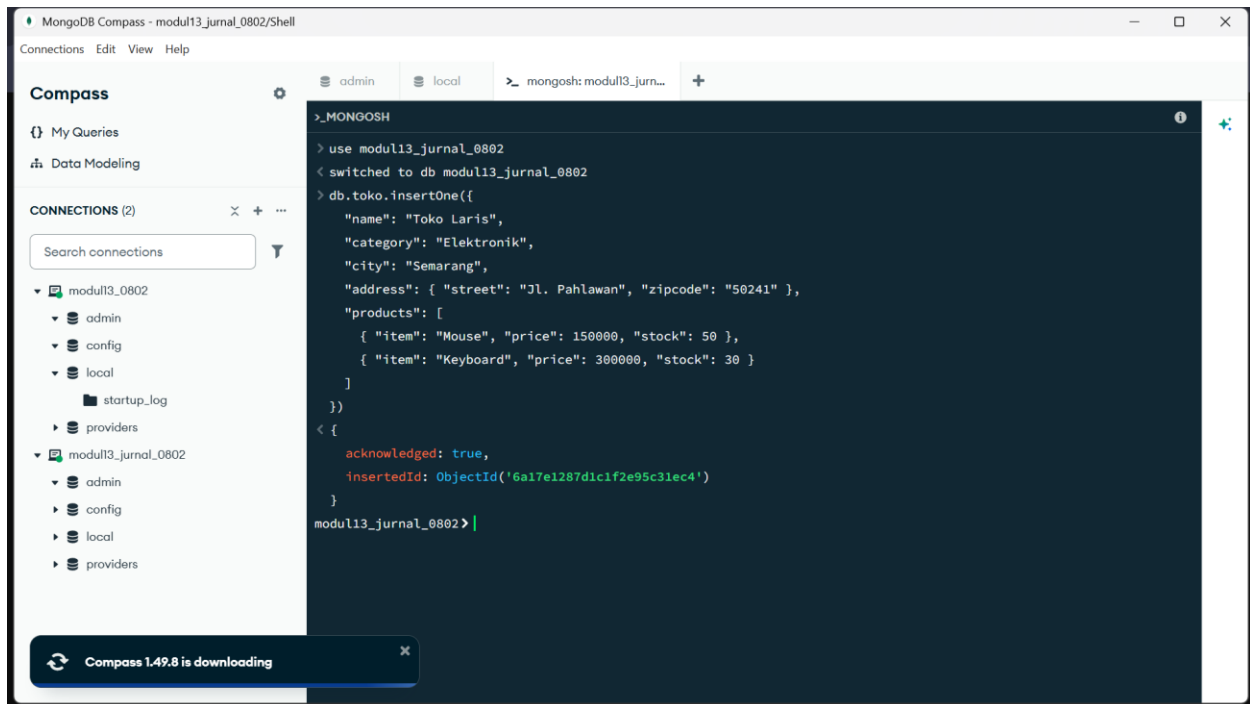
- category adalah Fashion
- city adalah Surabaya

Hint:

- Gunakan \$and atau tulis kedua kondisi dalam satu objek filter

JAWAB :

1.



2.



>_MONGOSH



```
}
{
  _id: ObjectId('6a17e1997d1c1f2e95c31ec7'),
  name: 'Toko Salim Bakery',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'Monitor LED',
      price: 7700000,
      stock: 11
    },
    {
      item: 'Airfryer Panasonic',
      price: 4850000,
      stock: 7
    }
  ]
}
```

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3.

```
>_MONGOSH
> db.toko.find({ "products.price": { $gt: 3000000 } })
< {
  _id: ObjectId('6a17e1997d1c1f2e95c31ec5'),
  name: 'Toko Berkah',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'TV',
      price: 3500000,
      stock: 10
    },
    {
      item: 'Kulkas',
      price: 2800000,
      stock: 5
    }
  ]
}
```

>_MONGOSH



```
_id: ObjectId('6a17e1997d1c1f2e95c31ec6'),
name: 'Toko Maju',
category: 'Elektronik',
city: 'Bandung',
address: {
  street: 'Jl. Asia Afrika',
  zipcode: '40111'
},
products: [
  {
    item: 'HP',
    price: 1500000,
    stock: 20
  },
  {
    item: 'Laptop',
    price: 8000000,
    stock: 3
  }
]
}
{
  _id: ObjectId('6a17e1997d1c1f2e95c31ec7'),
  name: 'Toko Salim Bakery',
```

>_MONGOSH



```
category: 'Elektronik',
city: 'Jakarta',
address: {
  street: 'Jl. Sudirman',
  zipcode: '10220'
},
products: [
  {
    item: 'Monitor LED',
    price: 7700000,
    stock: 11
  },
  {
    item: 'Airfryer Panasonic',
    price: 4850000,
    stock: 7
  }
]
}
{
  _id: ObjectId('6a17e1997d1c1f2e95c31ec8'),
  name: 'Toko Sudinotech',
  category: 'Elektronik',
  city: 'Surabaya',
```



```

>_MONGOSH
]
}
{
  _id: ObjectId('6a17e1997d1c1f2e95c31ec8'),
  name: 'Toko Sudinotech',
  category: 'Elektronik',
  city: 'Surabaya',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'RTX 55000',
      price: 8500000,
      stock: 20
    },
    {
      item: 'Set PC Gaming Radeon',
      price: 12000000,
      stock: 9
    }
  ]
}

```

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4.

```

> db.toko.updateOne(
  { "name": "Toko Berkah" },
  { $set: { "category": "Aksesoris" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}

```

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5.

```
> db.toko.aggregate([
  { $group: { _id: "$city", jumlah_toko: { $sum: 1 } } },
  { $sort: { jumlah_toko: -1 } }
])
< {
  _id: 'Jakarta',
  jumlah_toko: 2
}
{
  _id: 'Bandung',
  jumlah_toko: 1
}
{
  _id: 'Semarang',
  jumlah_toko: 1
}
{
  _id: 'Surabaya',
  jumlah_toko: 1
}
modul13_jurnal_0802> |
```

6.

```
> db.toko.aggregate([
  { $group: { _id: "$city", jumlah_toko: { $sum: 1 } } },
  { $sort: { jumlah_toko: -1 } }
])
< {
  _id: 'Jakarta',
  jumlah_toko: 2
}
{
  _id: 'Bandung',
  jumlah_toko: 1
}
{
  _id: 'Semarang',
  jumlah_toko: 1
}
{
  _id: 'Surabaya',
  jumlah_toko: 1
}
```

7.

```
> db.toko.updateOne(
  { "name": "Toko Maju" },
  { $push: { "products": { "item": "Powerbank", "price": 250000, "stock": 15 } } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
modul13_jurnal_0802>
```

8.

```
> db.toko.aggregate([
  { $unwind: "$products" },
  { $group: { _id: "$city", harga_tertinggi: { $max: "$products.price" } } }
])
< {
  _id: 'Jakarta',
  harga_tertinggi: 7700000
}
{
  _id: 'Semarang',
  harga_tertinggi: 300000
}
{
  _id: 'Bandung',
  harga_tertinggi: 8000000
}
{
  _id: 'Surabaya',
  harga_tertinggi: 12000000
}
modul13_jurnal_0802> |
```

9.

```
> db.toko.aggregate([
  { $match: { "category": "Elektronik" } },
  { $unwind: "$products" },
  { $group: { _id: "$city", total_stock: { $sum: "$products.stock" } } },
  { $sort: { total_stock: -1 } }
])
< {
  _id: 'Semarang',
  total_stock: 80
}
{
  _id: 'Bandung',
  total_stock: 38
}
{
  _id: 'Surabaya',
  total_stock: 29
}
{
  _id: 'Jakarta',
  total_stock: 18
}
modul13_jurnal_0802> |
```

10.

```
> db.toko.deleteMany({  
  "category": "Fashion",  
  "city": "Surabaya"  
})  
< {  
  acknowledged: true,  
  deletedCount: 0  
}  
modul13_jurnal_0802> |
```